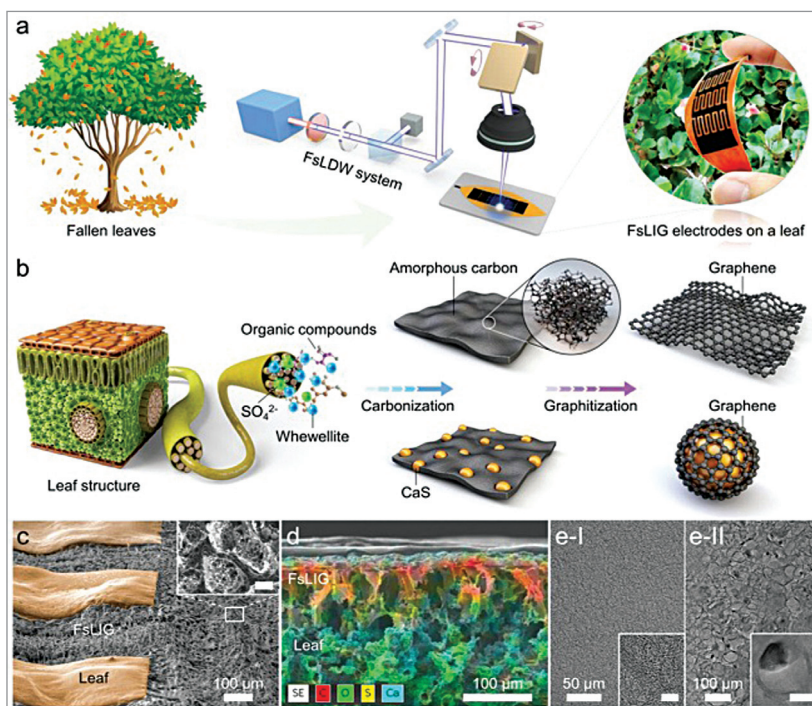


Eco-friendly micro-supercapacitors built with fallen leaves



A KAIST research team has developed graphene-inorganic-hybrid micro-supercapacitors made of fallen leaves using femtosecond laser direct writing. By irradiating femtosecond laser pulses on the leaves in ambient air, high electrical conductivity can be achieved. This one-step fabrication does not require any additional materials or pre-treatment. The technique could quickly and easily produce porous graphene electrodes at a low price. Potential applications include fabricating graphene micro-supercapacitors to power an LED and an electronic watch.

The schematic illustration of the production of femtosecond laser-induced graphene (Credit: <https://news.kaist.ac.kr>)

Versatile power transfer technology for wireless charging

Researchers at Aalto University have developed a power transfer technology that works regardless of the position and orientation of the transmitter and receiver. The key idea is to arrange the transmitters in a grid with the current in neighbouring transmitters running in opposite directions—for example, a clockwise loop in one transmitter and counter-clockwise loops in its neighbours. This creates a chessboard-like grid of ‘positive’ and ‘negative’ transmitting coils with a magnetic flux between them. A receiver above the grid of transmitters captures the magnetic flux between positive and negative transmitters, which generates an electric current to charge the device.

Wireless charging of a retail robot at a test site (Credit: www.aalto.fi/en)

